Supplement to "Measuring Explanation Agreement in Federated Intrusion Detection: A Noise-Aware Evaluation Protocol"

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All values are EED (RBO divergence minus the bootstrap noise floor, $p = 0.9$) unless stated; $\pm$ gives 95% t-intervals over seeds; * marks Holm-adjusted $p < 0.05$.
Code: `https://github.com/Sisty37/fedxai-ids-explanation-agreement`.

1 EED recipe (pseudocode)

```
input : SHAP matrices Phi_a, Phi_b (n_a x F, n_b x F) of ONE class, computed with the
        SAME shared background (100 flows, class-stratified from the reference set R)
s(Phi)        = mean(|Phi|, axis=0) / sum(mean(|Phi|, axis=0))      # importance vector
rank(s)       = features by decreasing s; ties (incl. exact zeros) by feature index
D(s, s')      = 1 - RBO_ext(rank(s), rank(s'); p = 0.9)
floor(Phi)    : draw B = 1000 resamples of the n rows of Phi with replacement
                (SHAP values reused, not recomputed); importance vector of each;
                draw 5000 random pairs (b, b'), b != b'; return mean D over the pairs
EED(a, b)     = D(s_a, s_b) - (floor(Phi_a) + floor(Phi_b)) / 2
different?    : D(s_a, s_b) > 95th percentile of the pooled within-bootstrap D values
classes with < 30 explained flows are skipped; explained flows: up to 200 per class,
uniformly sampled from R or from a client's test shard
```

2 CICIDS2017: tables and figures moved from the paper

Table S1: Data–model factorized analysis (per-class EED, mean $\pm$ 95% t-interval over ten seeds). Descriptive factorization, not a causal decomposition.

α	Term	Benign	DoS	DDoS	PortScan	BruteF.	Bot
1	data	0.00 ± 0.00	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.01	0.04 ± 0.05
	model	0.45 ± 0.03	0.55 ± 0.01	0.69 ± 0.01	0.75 ± 0.02	0.56 ± 0.03	0.65 ± 0.07
	combined	0.45 ± 0.03	0.56 ± 0.02	0.69 ± 0.01	0.74 ± 0.02	0.55 ± 0.03	0.65 ± 0.07
	interaction	0.00 ± 0.01	0.01 ± 0.01	-0.01 ± 0.01	-0.02 ± 0.01	0.00 ± 0.01	-0.04 ± 0.05
0.1	data	0.00 ± 0.01	0.01 ± 0.02	0.00 ± 0.01	0.01 ± 0.03	0.00 ± 0.01	0.03 ± 0.03
	model	0.59 ± 0.05	0.62 ± 0.06	0.71 ± 0.07	0.68 ± 0.04	0.55 ± 0.06	0.62 ± 0.10
	combined	0.59 ± 0.05	0.62 ± 0.06	0.71 ± 0.07	0.66 ± 0.05	0.56 ± 0.05	0.64 ± 0.10
	interaction	0.01 ± 0.02	-0.01 ± 0.02	0.00 ± 0.01	-0.03 ± 0.04	0.00 ± 0.02	-0.02 ± 0.03

Table S2: FedProx ($\mu = 0.01$) minus FedAvg, paired by seed: macro-F1 difference and per-class fidelity EED difference (positive = farther from centralized). Holm-corrected: * $p < .05$, ** $p < .01$, *** $p < .001$.

α	ΔMacro-F1	Benign	DoS	DDoS	PortScan	BruteF.	Bot
100	$+0.025^{*}$	$+0.52^{***}$	$+0.36^{***}$	$+0.20^{***}$	$+0.16^{***}$	$+0.01$	-0.10^{*}
5	$+0.017$	$+0.48^{***}$	$+0.39^{***}$	$+0.20^{***}$	$+0.18^{***}$	$+0.03$	-0.10^{*}
1	$+0.001$	$+0.24^{**}$	$+0.35^{***}$	$+0.22^{***}$	$+0.23^{***}$	$+0.05^{*}$	-0.15^{**}
0.5	$+0.013$	$+0.30^{***}$	$+0.40^{***}$	$+0.22^{***}$	$+0.26^{***}$	$+0.05$	-0.17^{**}
0.3	$+0.085^{*}$	$+0.30^{***}$	$+0.40^{***}$	$+0.26^{***}$	$+0.30^{***}$	$+0.07$	-0.11^{*}
0.1	$+0.152^{**}$	$+0.20$	$+0.27^{**}$	$+0.16^{***}$	$+0.19^{*}$	$+0.07$	-0.12

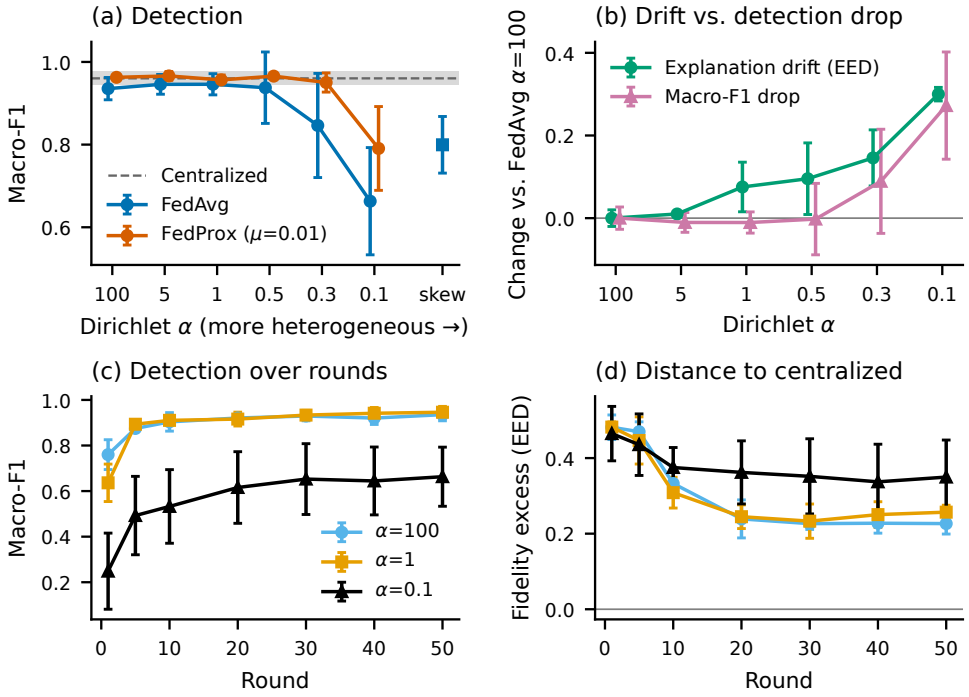


Figure S1: CICIDS2017, full version of the detection figure: (a) macro-F1 vs. heterogeneity; (b) explanation drift and macro-F1 drop relative to FedAvg at $\alpha = 100$ (shaded: $\alpha = 0.3$, drift significant but F1 drop not); (c,d) macro-F1 and fidelity excess over rounds.

Table S3: Full replication table (the paper shows a shortened version; EED unless stated; ranges over classes).

	CICIDS2017, MLP	CICIDS2017, CNN	UNSW-NB15, MLP
Seeds; macro-F1, central / $\alpha = 0.1$	10; 0.966 / 0.648	5; 0.961 / 0.537	10; 0.617 / 0.328
Client divergence from attack mix	82–93%	75–80%	69–83%
Local background (attacks, $\alpha = 1$)	0.30–0.64	0.43–0.71	0.17–0.35
Seed baseline	0.02–0.28	0.52–0.66	0.10–0.18
Federation excess, $\alpha = 100$ (sig.)	0.18–0.33 (6/6)	0.10–0.11 (2/6)	0.04–0.25 (9/9)
Drift, $\alpha = 0.1$ (class avg.)	0.28 ± 0.04	0.11 ± 0.03	0.26 ± 0.03
Drift sig., F1 drop not	$\alpha = 1$–0.3	–	$\alpha = 5$
Run-level ρ within α	0.23	–	0.52
FedProx: ΔF1; attributions	+0.15; farther	+0.05 ($\alpha = 100$); farther	+0.00; farther (3/9)

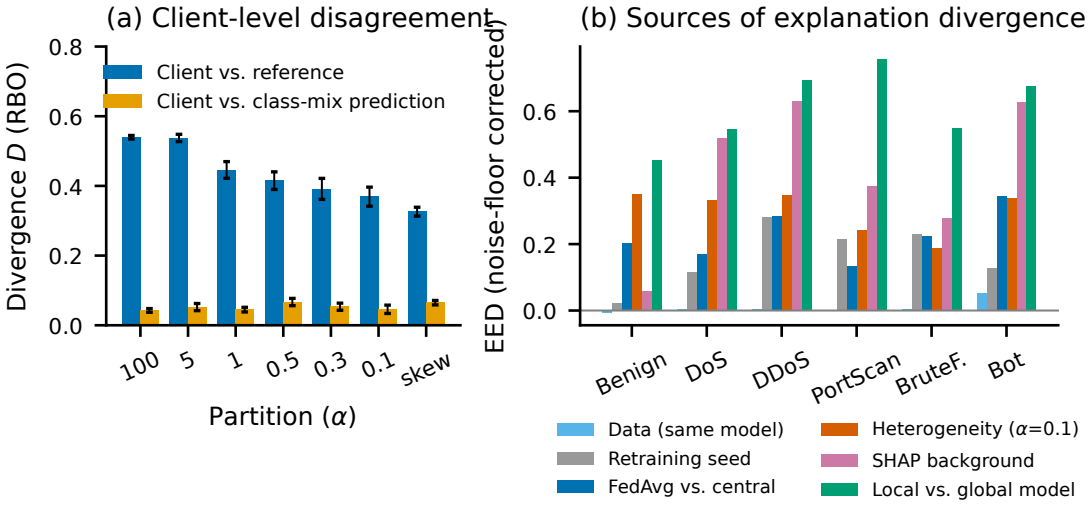


Figure S2: (a) Client-level divergence from the reference and from the class-mix prediction (same global model). (b) Per-class EED of each source; data, background and local-vs-global at $\alpha = 1$, heterogeneity at $\alpha = 0.1$.

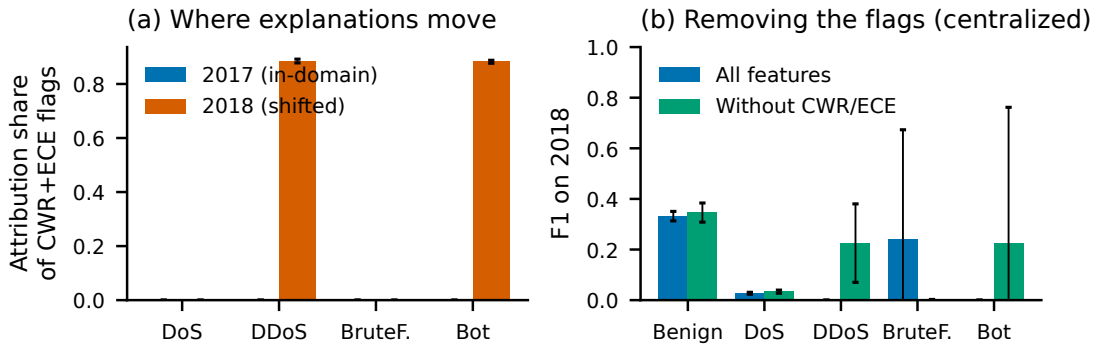


Figure S3: CICIDS2017→CSE-CIC-IDS2018. (a) Share of attribution on the CWR and ECE flag counts (centralized models). (b) 2018 F1 of centralized models trained with and without these features.

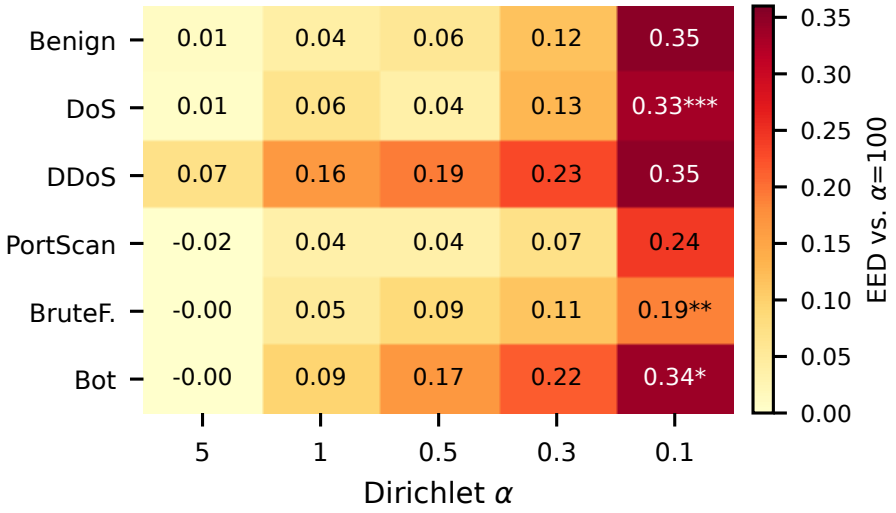


Figure S4: Heterogeneity drift per class and α (relative to FedAvg at $\alpha = 100$).

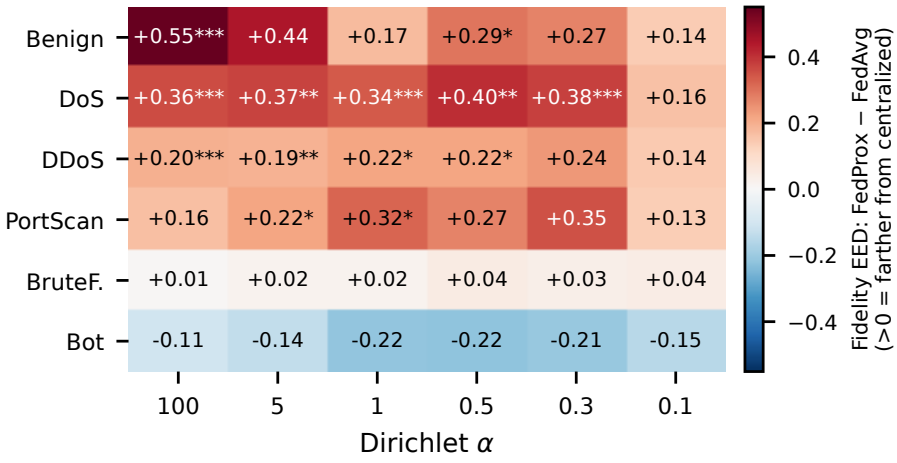


Figure S5: FedProx minus FedAvg fidelity difference per class and α.

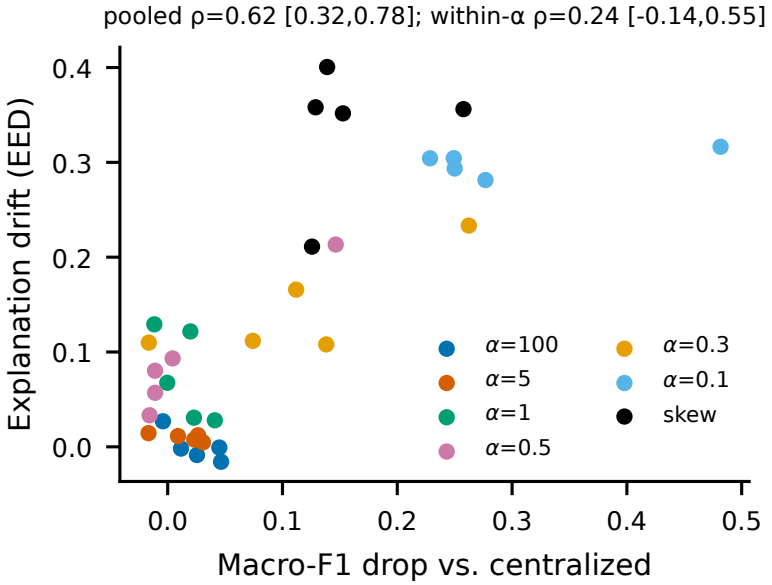


Figure S6: Run-level coupling of explanation drift and macro-F1 drop (70 FedAvg runs).

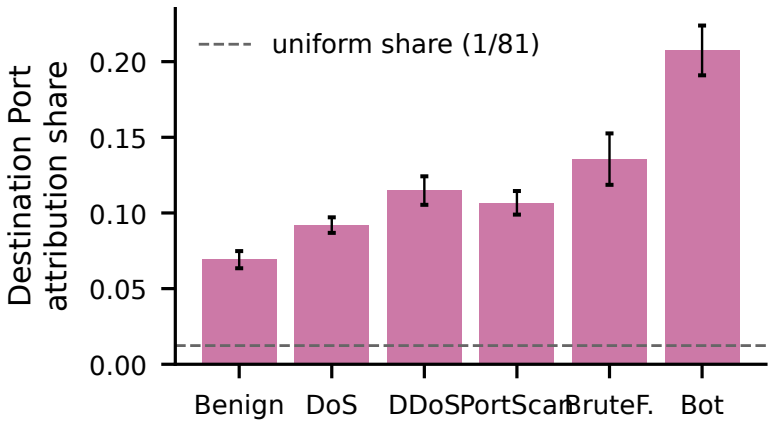


Figure S7: Attribution share of the destination port when it is kept.

3 CICIDS2017: sensitivity analyses

Table S4: Ranking weight: seed baseline per class; class-averaged federation excess ($\alpha = 100$) and drift ($\alpha = 0.1$); attack-mix share at $\alpha = 1$ / 0.1; classes with significant excess / drift (of 6); Spearman correlation of the 12 per-class effects with $p = 0.9$.

Divergence	Benign	DoS	DDoS	PortScan	BruteF.	Bot	Fed. excess	Drift	Mix share	Sig.	ρ
$p = 0.8$	0.00	0.13	0.33	0.32	0.23	0.17	0.26 ± 0.02	0.31 ± 0.05	92% / 86%	6/6	0.95
$p = 0.9$	0.02	0.13	0.28	0.25	0.23	0.15	0.24 ± 0.01	0.28 ± 0.04	89% / 85%	6/6	1.00
$p = 0.95$	0.03	0.11	0.23	0.18	0.20	0.13	0.20 ± 0.01	0.23 ± 0.03	86% / 84%	6/6	0.90
$p = 0.98$	0.03	0.08	0.14	0.11	0.13	0.09	0.12 ± 0.01	0.15 ± 0.02	84% / 83%	6/6	0.66
Spearman	0.02	0.03	0.06	0.03	0.05	0.04	0.07 ± 0.01	0.12 ± 0.03	87% / 88%	6/6	0.08

Table S5: Per-class federation excess ($\alpha = 100$, left) and drift ($\alpha = 0.1$, right) for each divergence.

Divergence	Benign	DoS	DDoS	PortScan	BruteF.	Bot	Benign	DoS	DDoS	PortScan	BruteF.	Bot
$p = 0.8$	0.19^*	0.22^*	0.37^*	0.18^*	0.21^*	0.41^*	0.31^*	0.32^*	0.37^*	0.28^*	0.18^*	0.40^*
$p = 0.9$	0.22^*	0.18^*	0.32^*	0.18^*	0.22^*	0.33^*	0.28^*	0.28^*	0.32^*	0.25^*	0.21^*	0.32^*
$p = 0.95$	0.20^*	0.15^*	0.25^*	0.17^*	0.18^*	0.25^*	0.24^*	0.23^*	0.26^*	0.21^*	0.20^*	0.25^*
$p = 0.98$	0.13^*	0.10^*	0.15^*	0.11^*	0.11^*	0.14^*	0.16^*	0.14^*	0.16^*	0.14^*	0.14^*	0.15^*
Spearman	0.08^*	0.05^*	0.09^*	0.10^*	0.06^*	0.06^*	0.13^*	0.10^*	0.12^*	0.16^*	0.13^*	0.10^*

Table S6: Shared background size: seed baseline per class, class-averaged federation excess and drift (all classes significant for every size).

Background	Benign	DoS	DDoS	PortScan	BruteF.	Bot	Fed. excess	Drift
25	0.03	0.21	0.22	0.21	0.23	0.11	0.25 ± 0.02	0.29 ± 0.04
50	0.01	0.14	0.15	0.13	0.15	0.09	0.36 ± 0.02	0.29 ± 0.05
100 (main)	0.02	0.13	0.28	0.25	0.23	0.15	0.24 ± 0.01	0.28 ± 0.04
200	0.01	0.16	0.28	0.21	0.22	0.18	0.25 ± 0.01	0.29 ± 0.05

Cost (NVIDIA GeForce RTX 4050 Laptop GPU): expected gradients (1000 samples) for the 1052 reference flows take 0.43 s per model; one noise floor ($B = 1000$, 5,000 pairs) takes 0.29 s per class; a client's per-class summary is $F + 1 = 81$ numbers (1944 bytes for six classes in float32).

Table S7: Same model (30 models), two different random 100-flow shared backgrounds: EED between the explanations.

Comparison	Benign	DoS	DDoS	PortScan	BruteF.	Bot
Draw B vs. main	-0.023 ± 0.009	0.107 ± 0.028	0.257 ± 0.054	0.084 ± 0.028	0.144 ± 0.030	0.178 ± 0.042
Draw C vs. main	-0.023 ± 0.010	0.048 ± 0.016	0.134 ± 0.037	0.027 ± 0.023	0.066 ± 0.015	0.072 ± 0.027

Table S8: Number of explained flows n: noise floor and EED between two centralized seeds (mean and SD over 5 seed pairs $\times$ 10 random subsets). Columns: floor | EED.

Class	$n = 10$		$n = 20$		$n = 30$		$n = 50$		$n = 100$		all R flows (n)	
Benign	0.214 (0.076)	-0.129 (0.080)	0.167 (0.063)	-0.077 (0.061)	0.147 (0.052)	-0.060 (0.053)	0.128 (0.036)	-0.048 (0.044)	0.098 (0.024)	-0.018 (0.022)	0.069	0.019 (200)
DoS	0.179 (0.066)	0.049 (0.092)	0.140 (0.045)	0.089 (0.053)	0.116 (0.031)	0.107 (0.058)	0.087 (0.022)	0.119 (0.056)	0.067 (0.016)	0.139 (0.052)	0.049	0.149 (200)
DDoS	0.156 (0.043)	0.248 (0.088)	0.129 (0.044)	0.284 (0.083)	0.114 (0.042)	0.299 (0.071)	0.087 (0.036)	0.319 (0.080)	0.065 (0.031)	0.333 (0.083)	0.039	0.340 (200)
PortScan	0.142 (0.040)	0.156 (0.113)	0.112 (0.029)	0.169 (0.085)	0.101 (0.029)	0.185 (0.093)	0.088 (0.026)	0.210 (0.087)	0.067 (0.025)	0.220 (0.052)	0.056	0.241 (167)
BruteForce	0.126 (0.021)	0.134 (0.044)	0.099 (0.016)	0.173 (0.045)	0.085 (0.017)	0.181 (0.046)	0.062 (0.014)	0.198 (0.049)	0.043 (0.014)	0.212 (0.045)	0.026	0.226 (200)
Bot	0.159 (0.074)	0.089 (0.094)	0.159 (0.045)	0.081 (0.067)	0.132 (0.044)	0.115 (0.064)	0.111 (0.033)	0.130 (0.054)	–	–	0.096	0.158 (74)

Table S9: Client–class cells (10 seeds $\times$ 10 clients $\times$ 7 classes) passing the minimum-flow threshold T.

Partition	$T = 10$	$T = 30$	$T = 50$	$T = 100$	max
$\alpha = 1$	524	434	400	355	700
$\alpha = 0.1$	296	257	240	209	700

Table S10: Bootstrap configuration: noise floor (Monte Carlo SD over 10 repetitions) for resamples B and number of pairs; centralized model, seed 0. Main setting: $B = 1000$, 5k pairs.

Class	$B = 200$, 1k	$B = 200$, 5k	$B = 200$, 20k	$B = 1000$, 1k	$B = 1000$, 5k	$B = 1000$, 20k	$B = 2000$, 1k	$B = 2000$, 5k	$B = 2000$, 20k
Benign	0.0721 (0.0019)	0.0721 (0.0020)	0.0719 (0.0019)	0.0726 (0.0010)	0.0723 (0.0009)	0.0723 (0.0009)	0.0721 (0.0011)	0.0723 (0.0008)	0.0723 (0.0007)
DoS	0.0404 (0.0031)	0.0404 (0.0026)	0.0404 (0.0025)	0.0394 (0.0018)	0.0398 (0.0011)	0.0397 (0.0012)	0.0392 (0.0016)	0.0390 (0.0013)	0.0392 (0.0013)
DDoS	0.0706 (0.0038)	0.0710 (0.0031)	0.0709 (0.0030)	0.0715 (0.0033)	0.0714 (0.0019)	0.0717 (0.0020)	0.0712 (0.0021)	0.0718 (0.0011)	0.0719 (0.0009)
PortScan	0.0343 (0.0007)	0.0344 (0.0009)	0.0343 (0.0009)	0.0342 (0.0005)	0.0343 (0.0005)	0.0343 (0.0004)	0.0345 (0.0006)	0.0342 (0.0003)	0.0343 (0.0004)
BruteForce	0.0223 (0.0013)	0.0223 (0.0011)	0.0222 (0.0011)	0.0226 (0.0005)	0.0225 (0.0005)	0.0227 (0.0005)	0.0226 (0.0005)	0.0226 (0.0004)	0.0227 (0.0003)
Bot	0.0758 (0.0031)	0.0758 (0.0033)	0.0759 (0.0029)	0.0771 (0.0020)	0.0768 (0.0012)	0.0764 (0.0010)	0.0765 (0.0013)	0.0766 (0.0011)	0.0766 (0.0008)

Table S11: KernelSHAP (1,000 samples, 50-flow background) vs. expected gradients on the same 20 flows per class; 5 centralized, 5 FedAvg $\alpha = 100$ and 5 FedAvg $\alpha = 0.1$ models. With 20 flows the floor is large, so seed baselines are noisy (negative for Benign).

Quantity	Benign	DoS	DDoS	PortScan	BruteF.	Bot
KernelSHAP, seed baseline	-0.22	0.11	0.05	0.15	0.11	0.10
KernelSHAP, federation excess	0.29 ± 0.02	0.06 ± 0.01	0.26 ± 0.04	0.18 ± 0.08	0.18 ± 0.02	0.14 ± 0.12
KernelSHAP, drift $\alpha = 0.1$	0.29 ± 0.24	0.31 ± 0.15	0.23 ± 0.10	0.33 ± 0.12	0.22 ± 0.12	0.26 ± 0.14
Exp. gradients, seed baseline	-0.19	0.08	0.21	0.15	0.18	0.12
Exp. gradients, federation excess	0.34 ± 0.01	0.15 ± 0.08	0.30 ± 0.03	0.18 ± 0.03	0.21 ± 0.04	0.32 ± 0.10
Exp. gradients, drift $\alpha = 0.1$	0.32 ± 0.16	0.29 ± 0.04	0.30 ± 0.13	0.27 ± 0.09	0.20 ± 0.05	0.31 ± 0.13
Rank agreement (Spearman)	0.84	0.65	0.48	0.62	0.62	0.60

4 UNSW-NB15 replication

UNSW-NB15 training- and testing-set CSVs: 257,673 flows; 94,928 duplicates and 11,506 label-conflict flows removed; 151,239 flows, 64 features (numeric features plus one-hot protocol, service and state, eight most frequent training values plus 'other'). Split, reference set, background, partitions, model, training and explanation settings as for CICIDS2017; ten seeds; FedAvg at six α, FedProx ($\mu = 0.01$) at $\alpha \in \{100, 0.1\}$.

Table S12: UNSW-NB15 per class: test flows, reference flows and F1 (macro-F1 in parentheses).

	Normal	Exploits	Fuzzers	Recon.	Generic	DoS	Shellc.	Analysis	Backdoor	Worms
Test flows	17,061	5,090	3,750	1,684	1,457	725	289	90	70	32
R flows	200	200	200	200	200	200	145	45	35	16
Centralized (0.617)	0.886	0.825	0.661	0.857	0.912	0.399	0.504	0.233	0.566	0.330
FedAvg $\alpha = 100$ (0.553)	0.867	0.806	0.625	0.771	0.895	0.300	0.489	0.216	0.330	0.234
FedAvg $\alpha = 1$ (0.531)	0.895	0.796	0.624	0.770	0.892	0.223	0.475	0.222	0.291	0.125
FedAvg $\alpha = 0.1$ (0.328)	0.886	0.526	0.226	0.412	0.773	0.137	0.219	0.056	0.041	0.005
FedProx $\alpha = 100$ (0.555)	0.867	0.807	0.625	0.769	0.894	0.295	0.488	0.216	0.328	0.257
FedProx $\alpha = 0.1$ (0.330)	0.883	0.507	0.228	0.387	0.866	0.127	0.189	0.059	0.047	0.006

Table S13: UNSW-NB15: seed baseline; fidelity excess over the seed baseline per α; heterogeneity drift relative to $\alpha = 100$. Last column: class average.

	Normal	Exploits	Fuzzers	Recon.	Generic	DoS	Shellc.	Analysis	Backdoor	Avg.
Seed baseline	0.14	0.11	0.10	0.18	0.16	0.10	0.17	0.14	0.12	–
Fidelity excess $\alpha = 100$	0.12^*	0.15^*	0.11^*	0.07^*	0.19^*	0.25^*	0.04^*	0.08^*	0.09^*	0.12 ± 0.01
Fidelity excess $\alpha = 5$	0.17^*	0.20^*	0.13^*	0.07^*	0.25^*	0.22^*	0.06^*	0.08^*	0.10^*	0.14 ± 0.01
Fidelity excess $\alpha = 1$	0.35^*	0.26^*	0.21^*	0.08^*	0.39^*	0.16^*	0.11^*	0.09^*	0.09^*	0.19 ± 0.01
Fidelity excess $\alpha = 0.5$	0.28^*	0.31^*	0.27^*	0.15^*	0.34^*	0.21^*	0.13^*	0.16^*	0.11^*	0.22 ± 0.02
Fidelity excess $\alpha = 0.3$	0.23^*	0.30^*	0.30^*	0.18^*	0.38^*	0.24^*	0.14^*	0.24^*	0.14^*	0.24 ± 0.02
Fidelity excess $\alpha = 0.1$	0.17^*	0.34^*	0.29^*	0.27^*	0.33^*	0.26^*	0.20^*	0.32^*	0.18^*	0.26 ± 0.03
Drift $\alpha = 5$	0.02^*	0.02	0.01	-0.00	0.11^*	0.04	0.00	0.00	0.00	0.02 ± 0.01
Drift $\alpha = 1$	0.29^*	0.20^*	0.13^*	0.05^*	0.27^*	0.17^*	0.04^*	0.07^*	0.07^*	0.14 ± 0.02
Drift $\alpha = 0.5$	0.32^*	0.28^*	0.24^*	0.14^*	0.29^*	0.23^*	0.08^*	0.17^*	0.10^*	0.21 ± 0.01
Drift $\alpha = 0.3$	0.31^*	0.26^*	0.29^*	0.19^*	0.32^*	0.21^*	0.10^*	0.29^*	0.13^*	0.23 ± 0.01
Drift $\alpha = 0.1$	0.22^*	0.30^*	0.31^*	0.30^*	0.34^*	0.20^*	0.15^*	0.39^*	0.14^*	0.26 ± 0.03

Table S14: UNSW-NB15: class-averaged drift vs. macro-F1 drop relative to FedAvg at $\alpha = 100$.

Partition	Drift	F1 drop
$\alpha = 5$	$0.022 \pm 0.009^*$	-0.000 ± 0.004
$\alpha = 1$	$0.143 \pm 0.020^*$	$0.022 \pm 0.018^*$
$\alpha = 0.5$	$0.206 \pm 0.014^*$	$0.063 \pm 0.021^*$
$\alpha = 0.3$	$0.232 \pm 0.015^*$	$0.093 \pm 0.037^*$
$\alpha = 0.1$	$0.261 \pm 0.026^*$	$0.225 \pm 0.060^*$
Run-level Spearman (drift vs. F1 drop, 50 runs): raw 0.88, within α 0.52		

Table S15: UNSW-NB15: FedProx ($\mu = 0.01$) minus FedAvg, paired by seed: macro-F1 and per-class fidelity EED (positive = farther from centralized).

	ΔF1	Normal	Exploits	Fuzzers	Recon.	Generic	DoS	Shellc.	Analysis	Backdoor
$\alpha = 100$	+0.001	-0.03	-0.02*	-0.01	+0.01*	-0.04*	+0.01	+0.01	+0.00	+0.01
$\alpha = 0.1$	+0.002	+0.18*	+0.01	+0.04	-0.01	+0.00	+0.00	+0.16*	-0.08	+0.16*

Table S16: UNSW-NB15 class mix (same global model): client divergence from the reference and from the class-mix prediction, share explained, and per-class client-vs-reference EED.

	D vs. R	D vs. mix	Share	Normal	Exploits	Fuzzers	Recon.	Generic	DoS	Shellc.	Analysis	Backdoor
$\alpha = 100$	0.208	0.037	82%	0.012	0.000	-0.002	-0.001	-0.007	-0.004	0.007	–	–
$\alpha = 5$	0.196	0.034	83%	0.013	0.003	-0.006	0.001	0.008	0.005	0.002	–	–
$\alpha = 1$	0.235	0.056	76%	0.018	0.005	-0.005	-0.001	0.003	0.015	0.000	0.020	0.023
$\alpha = 0.5$	0.213	0.060	72%	0.012	0.006	-0.001	0.002	0.001	-0.001	0.001	0.061	-0.002
$\alpha = 0.3$	0.191	0.053	72%	0.009	0.002	0.004	0.001	0.003	-0.014	0.006	0.021	0.010
$\alpha = 0.1$	0.198	0.061	69%	0.032	0.009	-0.001	-0.001	0.009	0.000	0.003	0.010	-0.006

Table S17: UNSW-NB15, $\alpha = 1$: effect of a client-local background (number of client cells in parentheses; Analysis and Backdoor have few cells and are not summarized in the paper).

	Normal	Exploits	Fuzzers	Recon.	Generic	DoS	Shellc.	Analysis	Backdoor
Local vs. shared (same flows)	0.16 (100)	0.23 (96)	0.35 (96)	0.26 (87)	0.17 (83)	0.18 (72)	0.21 (35)	0.39 (5)	0.32 (1)
Client (local) vs. R (shared)	0.19	0.26	0.35	0.26	0.17	0.22	0.22	0.37	0.31